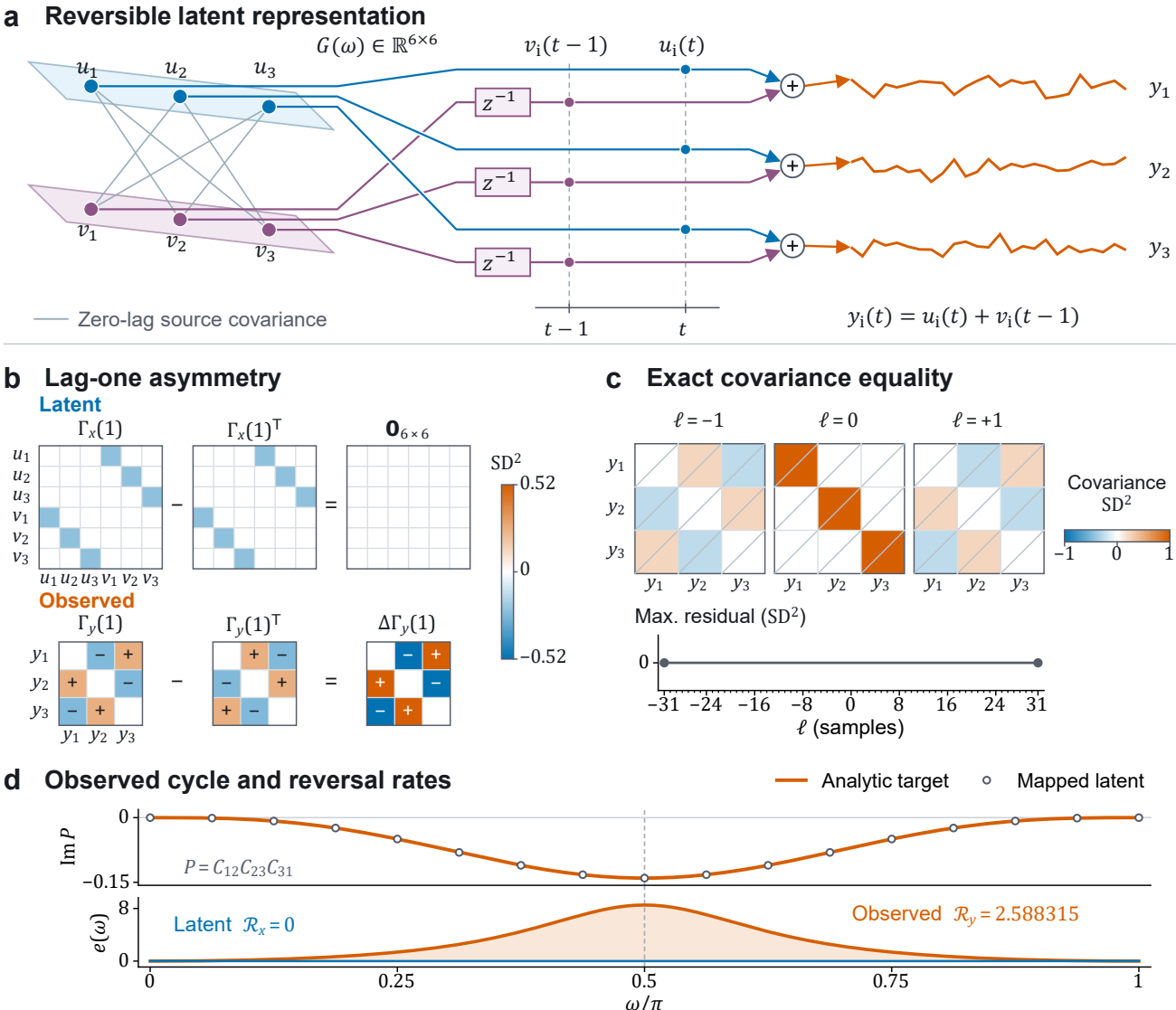

a
Reversible latent representation
$G(\omega) \in \mathbb{R}^{6\times6}$
$v_i(t-1)$
$u_i(t)$
u_1 u_2 u_3
v_1 v_2 v_3
z^{-1}
y_1 y_2 y_3
Zero-lag source covariance
$t-1$ t
$y_i(t) = u_i(t) + v_i(t-1)$
b
Lag-one asymmetry
Latent
$\Gamma_x(1)$ $\Gamma_x(1)^\top$ $\mathbf{0}_{6\times6}$
u_1 u_2 u_3 v_1 v_2 v_3
SD²
0.52
0
−0.52
Observed
$\Gamma_y(1)$ $\Gamma_y(1)^\top$ $\Delta\Gamma_y(1)$
y_1 y_2 y_3
c
Exact covariance equality
$\ell = -1$ $\ell = 0$ $\ell = +1$
Covariance SD²
−1 0 1
Max. residual (SD²)
0
−31 −24 −16 −8 0 8 16 24 31
ℓ (samples)
d
Observed cycle and reversal rates
Analytic target
Mapped latent
Im P
0
−0.15
$P = C_{12}C_{23}C_{31}$
$e(\omega)$
8
0
Latent $\mathcal{R}_x = 0$
Observed $\mathcal{R}_y = 2.588315$
0 0.25 0.5 0.75 1
ω/π